Stack Homework 1 Calculus 1

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4. $\left|2x^2 - \frac{331}{160}\right| < \frac{389}{160}$

$-\frac{389}{160} < 2x^2 - \frac{331}{160} < \frac{389}{160}$

$-\frac{58}{160} < 2x^2 < \frac{720}{160}$

$-\frac{29}{160} < x^2 < \frac{9}{4}$

$0 \leq x < \frac{3}{2}$

3. $1+2+3+\ldots+143 = ?$

$= \frac{n(a_1+a_n)}{2} = \frac{143 \cdot 144}{2} = 143 \cdot$

2. 1 4 2 2 2

2 4 4+2 2 3 2+1

3 4 10 6 8 2 5 3+2

4 14 10 16 6 14 8 10 2 9 5+4

$\#n+1 = 2n-1$ $\#n = n+2^{n-1}$

$1580 = 2 \cdot 1579 - 1 = 3157$

1. 1 4 $S_1 = 1+4$ 5

1 5 4 $S_2 = 1+4+(1+4)$ 10

1 6 9 4 $S_3 = 1+4+((1+4)+1)+((1+4)+4)$ 20

1 7 15 13 4 $S_4 = 40$

1 8 22 28 17 4 $S_5 = 5+25+50 = 80$

1 9 30 50 45 21 4 $S_6 = 5+30+75+50 = 160$

$S_n = 5 \cdot 2^{n-1}$

$S_{1583} = 5 \cdot 2^{1582}$